

ARJUN KRISHNAMURTHY

Senior Java Developer

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PROFESSIONAL SUMMARY

- 10+ years of experience in Java/J2EE enterprise application development, microservices architecture, and cloud-native AWS solutions across insurance, banking, retail, telecom, and logistics domains.
- Designed and developed enterprise-grade microservices using **Spring Boot, Spring Cloud, and Java 11/17**, deploying on **AWS (ECS, EKS, Lambda, EC2)**, with RESTful and event-driven APIs serving millions of transactions daily across distributed teams.
- Built and maintained production-grade AWS data pipelines using **AWS Glue, AWS Kinesis, Apache Kafka, and Apache Spark**, leveraging S3, EventBridge, Lambda, and RDS to ingest, transform, and deliver large-scale enterprise datasets.
- Implemented robust CI/CD pipelines using **Jenkins, Docker, Kubernetes (EKS), and AWS CDK**, enforcing SOX, PCI-DSS, HIPAA, and FedRAMP compliance standards in regulated enterprise environments.
- Expert in ORM frameworks (**Hibernate, Spring Data JPA**), working with MySQL, PostgreSQL, Oracle, AWS RDS Aurora, and DynamoDB to design optimized relational and NoSQL schemas for high-throughput applications.
- Developed event-driven microservices using **Apache Kafka, AWS SQS, AWS SNS, and AWS EventBridge**, enabling real-time data ingestion, processing, and asynchronous inter-service communication at scale.
- Proficient in observability using **AWS CloudWatch, Prometheus, Grafana, and Splunk**, implementing distributed tracing, centralized logging, and automated alerting across production-grade Java microservices environments.
- Led containerization initiatives using **Docker and Kubernetes**, managing Helm charts, resource quotas, HPA policies, and multi-AZ deployments for zero-downtime releases and high availability.
- Strong hands-on experience with **AWS CDK, CloudFormation, and Terraform** for infrastructure-as-code provisioning of VPCs, IAM roles, EKS clusters, RDS instances, and API Gateway endpoints.
- Strong background in Agile/Scrum methodologies, conducting sprint planning, code reviews, architectural design sessions, and mentoring junior developers across globally distributed teams.

TECHNICAL SKILLS

Core Java & Frameworks: Java 8/11/17, Spring Boot, Spring MVC, Spring Cloud, Spring Security, Spring Data JPA, Hibernate, Microservices, REST APIs, GraphQL, gRPC, Multithreading, Design Patterns, JUnit 5, Mockito, TestContainers

Databases & Tools: MySQL, PostgreSQL, Oracle 12c/11g, MS SQL Server, AWS RDS Aurora, Amazon DynamoDB, MongoDB, Redis, Cassandra, Amazon Redshift

Cloud – AWS: EC2, S3, Lambda, ECS, EKS, RDS, DynamoDB, API Gateway, SQS, SNS, EventBridge, Kinesis, Glue, CloudWatch, CloudTrail, CDK, CloudFormation, Secrets Manager, IAM, VPC, WAF, ElastiCache, SageMaker, Step Functions, Athena

Big Data & Streaming: Apache Kafka, Apache Spark, Spark Streaming, AWS Kinesis, AWS Glue, Apache Airflow, Hive, HDFS, Flink, Yarn

DevOps & CI/CD: Jenkins, Docker, Kubernetes, Helm, AWS CDK, Terraform, Git, GitHub, GitLab, SonarQube, JFrog Artifactory, Maven, Gradle

Monitoring & Observability: AWS CloudWatch, Prometheus, Grafana, Splunk, ELK Stack, Datadog, AWS X-Ray, PagerDuty

Integration & Messaging: Apache Kafka, AWS SQS, AWS SNS, AWS EventBridge, RabbitMQ, MQ Series, REST, SOAP, GraphQL

Languages & Scripting: Java, Python, Groovy, SQL, Bash, TypeScript

Security & Compliance: OAuth2, JWT, Spring Security, AWS IAM, AWS WAF, AWS KMS, SSL/TLS, OWASP, PCI-DSS, HIPAA, SOX, FedRAMP

Reporting & BI: Power BI, Tableau, Amazon QuickSight, Splunk

Methodologies: Agile/Scrum, SAFe, Waterfall, SDLC, TDD, BDD, Domain-Driven Design

PROFESSIONAL EXPERIENCE

Aetna (CVS Health): September 2023 – Current (Location: Hartford, CT (Remote))

Role: Senior Java Developer

Responsibilities:

- Architected and developed HIPAA-compliant **Spring Boot microservices** deployed on **AWS ECS and EKS**, delivering RESTful APIs for member eligibility verification, claims adjudication, and provider network management, handling 2M+ daily transactions.
- Built event-driven claims processing pipelines using **Apache Kafka and AWS Kinesis**, integrating Spring Boot consumers with AWS Lambda triggers and S3 event notifications to process real-time insurance claim submissions with sub-second latency.
- Designed and deployed **AWS Step Functions workflows** orchestrating multi-step prior authorization processes, integrating Java Lambda functions with DynamoDB state management and SNS notifications for clinicians and patients.
- Implemented **AWS API Gateway with Spring Security OAuth2 and JWT**, securing member-facing and provider-facing APIs with fine-grained IAM role policies, Cognito user pools, and WAF rate limiting for HIPAA-compliant access control.
- Developed **Spring Data JPA and Hibernate ORM layers** against AWS RDS Aurora PostgreSQL clusters, implementing read replicas, connection pooling via HikariCP, and query optimization reducing average API response time from 850ms to 140ms.
- Built real-time member analytics pipelines using **AWS Glue ETL jobs and Apache Spark**, transforming HL7 FHIR clinical records from S3 into Redshift data marts powering Power BI dashboards for population health management.
- Engineered **AWS CDK infrastructure-as-code** in TypeScript provisioning EKS clusters, RDS Aurora, ElastiCache Redis, and API Gateway endpoints, automating blue/green deployments with CodeDeploy and enforcing HIPAA/HITECH compliance guardrails.
- Implemented distributed tracing and observability using **AWS X-Ray, CloudWatch Container Insights, and Datadog APM**, creating custom dashboards tracking claims processing latency, error rates, and SLA adherence across 30+ microservices.
- Led migration of legacy monolithic claims system to **Java 17 Spring Boot microservices** on EKS, decomposing 400K LOC monolith into 18 domain-driven services with Kafka event sourcing, reducing deployment frequency from monthly to daily.
- Integrated **SonarQube and OWASP Dependency-Check** into Jenkins CI pipelines for Java code quality gates, achieving 85% unit test coverage and eliminating critical CVEs before production deployment across 18 microservices.
- Collaborated with clinical informatics teams to implement **HL7 FHIR R4 Java APIs** using HAPI FHIR library, enabling interoperability across Epic EHR, pharmacy benefit managers, and lab systems under CMS Interoperability Rule compliance.

Environment: *Java 17, Spring Boot, Spring Cloud, Spring Security, Hibernate, AWS ECS, AWS EKS, AWS Lambda, AWS API Gateway, AWS Kinesis, Apache Kafka, AWS Glue, Apache Spark, AWS RDS Aurora, DynamoDB, AWS Step Functions, AWS CDK, Jenkins, Docker, Kubernetes, SonarQube, Datadog, AWS X-Ray, Power BI, Redshift*

Wells Fargo: July 2020 – August 2023 (Location: Charlotte, NC)

Role: Senior Java Developer

Responsibilities:

- Designed and deployed **Spring Boot and Spring Cloud microservices** for retail banking, mortgage origination, and credit card platforms on **AWS EKS**, processing 5M+ daily financial transactions with PCI-DSS and SOX-compliant audit logging.
- Built high-throughput **Apache Kafka event streaming pipelines** for real-time fraud detection, integrating Java consumers with AWS Lambda, AWS SageMaker ML models, and DynamoDB to flag suspicious transactions within 200ms of occurrence.
- Developed **Java Spring Batch jobs** for nightly ACH settlement processing, interest calculation, and regulatory reporting, running on AWS ECS Fargate with S3-triggered execution, processing 10M+ records per batch with checkpointing and retry logic.
- Implemented **AWS Secrets Manager and AWS KMS integration** in Spring Boot services for database credential rotation, encryption key management, and PII field-level encryption in customer account and transaction data stores.
- Engineered **API Gateway and Spring Security OAuth2** authentication flows for digital banking APIs, implementing MFA, session management, and token refresh mechanisms serving 8M active digital banking customers with 99.99% availability SLA.
- Architected **AWS CDK TypeScript infrastructure templates** provisioning multi-AZ RDS Aurora MySQL clusters, ElastiCache Redis for session caching, EKS node groups, and WAF rules for loan origination and mortgage platforms under PCI-DSS compliance.
- Built **real-time data pipelines using AWS Kinesis Data Streams and Firehose**, ingesting ATM and POS transaction events into S3 data lake, transforming with Glue Spark jobs, and loading into Redshift for risk analytics and regulatory capital reporting.
- Implemented **distributed transaction management using Saga pattern** with Kafka for fund transfer microservices, ensuring eventual consistency across account, ledger, and notification services without two-phase commit overhead.
- Built observability using **Prometheus, Grafana, and AWS CloudWatch**, implementing custom JVM metrics, Kafka consumer lag dashboards, and PagerDuty alerting for P1 incident response with mean-time-to-detect under 3 minutes.
- **Led Java 11 to Java 17 migration and Spring Boot 2.x to 3.x upgrade** across 25 services, modernizing to virtual threads (Project Loom), updating dependency chains, and enforcing stricter SonarQube quality gates reducing technical debt by 40%.

Environment: *Java 11/17, Spring Boot, Spring Cloud, Spring Batch, Spring Security, Hibernate, AWS EKS, AWS Lambda, Apache Kafka, AWS Kinesis, AWS Glue, Apache Spark, AWS RDS Aurora, DynamoDB, ElastiCache, AWS CDK, AWS KMS, AWS Secrets Manager, Jenkins, Docker, Kubernetes, Prometheus, Grafana, PagerDuty, Redshift, SonarQube*

Target Corporation: March 2018 – June 2020 (Location: Minneapolis, MN)

Role: Java Developer

Responsibilities:

- Developed **Spring Boot microservices** for inventory management, order orchestration, and fulfillment routing on AWS ECS, handling Black Friday peak loads of 50K+ requests/second with auto-scaling policies and circuit breakers via Resilience4j.
- Built **Apache Kafka event-driven order pipeline** integrating Java producers/consumers across order capture, payment processing, warehouse management, and last-mile delivery services, enabling real-time order tracking for 40M+ customers.

- Designed **Spring Data JPA repositories** against MySQL RDS with read replicas, implementing second-level Hibernate caching with Redis ElastiCache, reducing database load by 60% during peak promotional events.
- Implemented **AWS Lambda functions in Java** for serverless price calculation, promotional discount application, and loyalty point accrual, triggered by API Gateway and SQS events with sub-100ms cold start optimization.
- Built **AWS Glue and Apache Spark ETL jobs** processing daily sales, inventory, and supplier data from S3, transforming and loading into Redshift data warehouse powering QuickSight dashboards for category managers and supply chain teams.
- Developed **GraphQL APIs using Java Spring Boot** for the Target.com product catalog and search platform, federating data across product, inventory, pricing, and review microservices with DataLoader batching to eliminate N+1 query performance issues.
- Implemented **blue/green deployments using AWS CodeDeploy and ECS**, enabling zero-downtime releases for critical order management services and cutting rollback time from 45 minutes to under 2 minutes during holiday peaks.
- Collaborated with data science teams to integrate **AWS SageMaker ML models** into Java product recommendation and demand forecasting services, deploying models as SageMaker endpoints consumed via Spring Boot REST client with fallback caching.

Environment: *Java 11, Spring Boot, Spring Cloud, Spring Data JPA, Hibernate, Apache Kafka, AWS ECS, AWS Lambda, AWS Glue, Apache Spark, MySQL RDS, ElastiCache Redis, AWS Redshift, AWS CDK, Jenkins, Docker, Kubernetes, SonarQube, Splunk, AWS CloudWatch, Amazon QuickSight, GraphQL*

Verizon Communications: January 2016 – February 2018 (Location: Basking Ridge, NJ)

Role: Java Developer

Responsibilities:

- Built **Spring Boot RESTful microservices** for network provisioning, subscriber billing, and service activation on AWS EC2 and ECS, processing real-time telecom event data for 100M+ wireless subscribers across LTE and 5G networks.
- Developed **Java Kafka consumers and producers** for CDR (Call Detail Record) streaming pipeline, ingesting 500M+ daily records from network probes into S3 data lake via Kinesis Firehose for billing reconciliation and fraud analytics.
- Implemented **Spring Batch processing framework** for monthly invoice generation and revenue assurance workflows, running on AWS Fargate with partitioned job execution processing 80M customer accounts across billing cycles.
- Designed **Oracle and PostgreSQL RDS schemas** using Hibernate ORM for subscriber profile management, plan entitlement, and usage tracking, optimizing complex joins and partitioned tables for sub-second query performance on billion-row datasets.
- Built **AWS API Gateway and Spring Security** integration for B2B partner APIs, implementing mutual TLS authentication, OAuth2 client credentials flow, and rate limiting for enterprise account management portals serving Fortune 500 customers.
- Implemented **ELK Stack (Elasticsearch, Logstash, Kibana)** for centralized log aggregation across telecom microservices, creating Kibana dashboards tracking API error rates, network provisioning latency, and billing cycle completion rates.
- Built **AWS SQS-based asynchronous workflows** for service order management, decoupling order capture from network provisioning and equipment shipping systems, improving order completion rates and enabling graceful degradation during downstream outages.

Environment: *Java 8/11, Spring Boot, Spring Batch, Spring Security, Hibernate, AWS EC2, AWS ECS, AWS Lambda, Apache Kafka, AWS Kinesis, Oracle RDS, PostgreSQL RDS, AWS CDK, CloudFormation, Jenkins, Docker, Kubernetes, ELK Stack, AWS CloudWatch, SonarQube, Maven*

Cognizant Technology Solutions (Client: UPS): June 2014 – December 2015 (Location: Atlanta, GA)

Role: Java Developer

Responsibilities:

- Developed **Java Spring MVC and Spring Boot applications** for shipment tracking, route optimization, and customs documentation processing, integrating with UPS logistics APIs and third-party carrier systems handling 20M+ daily package events.
- Built **AWS S3 and Lambda-based document processing pipelines** for commercial invoice, bill of lading, and customs declaration digitization, using Java Textract integration to extract structured data from scanned shipping documents.
- Designed **Hibernate ORM data access layers** against Oracle 12c databases for shipment lifecycle management, implementing optimistic locking, batch inserts, and connection pooling to support 10K+ concurrent warehouse terminal sessions.
- Implemented **Apache ActiveMQ and AWS SQS messaging** for decoupling package scan events from tracking status updates, enabling real-time customer notifications via SNS for delivery milestones across ground, air, and international shipments.
- Built **ETL pipelines using Java and AWS Glue**, transforming operational shipment data into analytical datasets in Redshift for Business Intelligence reporting on network capacity planning and on-time delivery performance.
- Configured **Jenkins CI/CD pipelines** with Maven builds, JUnit/Mockito test execution, SonarQube quality gates, and automated deployment to AWS EC2 Auto Scaling groups, reducing release cycle from bi-weekly to weekly.

Environment: *Java 7/8, Spring Boot, Spring MVC, Spring Batch, Hibernate, Oracle 12c, AWS EC2, AWS S3, AWS Lambda, AWS SQS, AWS SNS, AWS Glue, Amazon Redshift, Redis, Apache ActiveMQ, Jenkins, Maven, SonarQube, JUnit, Mockito*

CERTIFICATIONS:

AWS Certified Solutions Architect – Professional – Amazon Web Services (AWS) | Issued: March 2023 | Credential ID: AWS-SAP-2023-AK7821

AWS Certified Developer – Associate – Amazon Web Services (AWS) | Issued: August 2021 | Credential ID: AWS-DVA-2021-AK4412

AWS Certified Cloud Practitioner – Amazon Web Services (AWS) | Issued: January 2020 | Credential ID: AWS-CCP-2020-AK1103

Oracle Certified Professional: Java SE 11 Developer – Oracle Corporation | Issued: November 2022 | Credential ID: OCP-SE11-2022-9934K

Oracle Certified Associate: Java SE 8 Programmer – Oracle Corporation | Issued: June 2018 | Credential ID: OCA-SE8-2018-7723J

Certified Kubernetes Application Developer (CKAD) – Cloud Native Computing Foundation (CNCF) | Issued: May 2022 | Credential ID: CKAD-2022-ARK-00341

Spring Professional Certification – VMware (Pivotal) | Issued: September 2020 | Credential ID: SPR-PROF-2020-VM8812

HashiCorp Certified: Terraform Associate – HashiCorp | Issued: February 2023 | Credential ID: HCTA-2023-001-ARK77

Confluent Certified Developer for Apache Kafka – Confluent Inc. | Issued: July 2022 | Credential ID: CCDAK-2022-CF5541

Microsoft Certified: Azure Fundamentals (AZ-900) – Microsoft | Issued: April 2021 | Credential ID: AZ900-2021-MS-9981A

EDUCATION:

Bachelor of Technology in Computer Science & Engineering – Jawaharlal Nehru Technological University (JNTU), Hyderabad, 2010–2014